ADALIMUMAB

Screen for TB & hepatitis B

Serious infections

CHEMICAL NAME Recombinant fully human IgG1κ monoclonal antibody against tumour necrosis factor-alpha (TNF-α)

CLASS Biologic — TNF-α inhibitor (monoclonal antibody)

TRADE NAMES Humira · Amgevita · Imraldi · Hyrimoz (biosimilars)

THERAPEUTIC USES / INDICATIONS

- Rheumatoid arthritis, psoriatic arthritis, ankylosing spondylitis, juvenile idiopathic arthritis.
- Crohn's disease, ulcerative colitis.
- Plaque psoriasis; hidradenitis suppurativa; non-infectious uveitis.

ADMINISTRATION

- **SC** injection (prefilled pen/syringe) into thigh or abdomen; rotate sites.
- Allow pen to reach room temperature (15–30 min); store in fridge.
- Usually with methotrexate in RA (reduces antibody formation).

DOSAGES

- **RA/PsA/AS:** 40 mg every 2 weeks SC.
- **Crohn's/UC:** 160 mg week 0, 80 mg week 2, then 40 mg every 2 weeks.
- **Psoriasis:** 80 mg week 0, then 40 mg every other week from week 1.
- **Children:** weight-based.

PHARMACOKINETICS

- **Onset:** weeks (RA response by 12 weeks).
- Bioavailability ~64% SC; catabolised like IgG.
- **t½:** ~2 weeks.

PHARMACODYNAMICS

- Binds and neutralises **soluble and membrane-bound TNF-α** → ↓ inflammatory cytokines, adhesion molecules, joint destruction.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Injection-site reactions.
- Upper respiratory infections, headache.
- Rash, nausea.

ADVERSE EFFECTS (SERIOUS)

- Serious infections and TB reactivation; opportunistic infections; hepatitis B reactivation.
- Lymphoma and other malignancies; non-melanoma skin cancer.
- Demyelinating disease (MS-like); heart failure worsening.
- Drug-induced lupus; cytopenias.

CONTRAINDICATIONS & CAUTIONS

- Active TB or other severe infections (sepsis, opportunistic).
- Moderate–severe heart failure (NYHA III/IV).
- Live vaccines during treatment.

KEY INTERACTIONS

- **Live vaccines:** contraindicated.
- **Other biologics (anakinra, abatacept):** ↑ infection — avoid combination.

NURSING MANAGEMENT

- Screen before starting: TB (IGRA/chest X-ray), hepatitis B/C, HIV; update vaccinations (non-live).
- Teach injection technique and storage.
- Monitor for infection at every visit; hold during serious infection.
- Assess disease activity scores.

PATIENT EDUCATION

- Store in fridge; let it warm before injecting; rotate sites.
- Report fever, cough, weight loss, night sweats (TB) or any infection early.
- Avoid live vaccines; inform doctors/dentists of treatment.
- Use sun protection; report new skin lesions.

KEY POINTS

- Anti-TNF-α monoclonal antibody.
- Screen for TB and hepatitis B first.
- 40 mg SC every 2 weeks.
- Serious infections; no live vaccines.